AI-Based Stroke Phase Detection and Symmetry Analysis in Competitive Swimming Footage: A Proof of Concept

Daniel Pace

Supervisor: Dr. Vanessa Camilleri

14th May 2026

Submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Information Technology (Honours) (Artificial Intelligence).

L-Università ta' Malta
Faculty of Information &
Communication Technology

Abstract

Computer vision analysis in aquatic environments presents unique challenges due to light refraction, surface turbulence, and occlusion, which hinder the accuracy of standard motion tracking systems. This project presents a non-intrusive, AI-based proof-of-concept system for extracting swimming kinematics from uncalibrated, handheld poolside video footage. The core contribution is a novel methodology for dynamic pixel-to-meter scaling using custom 3D-printed underwater markers as calibration objects, addressing the limitations of fixed-camera setups across varying pool environments. A Model-in-the-Loop annotation pipeline, combining SAM 2 temporal segmentation with statistical outlier filtering and human verification, was developed to construct a domain-specific training dataset for a fine-tuned YOLO26 marker detector. Ego-motion compensation, ViTPose-based hip localisation, and Kalman filter smoothing were combined to produce absolute positional and velocity profiles from raw tracking output. Evaluation on a 50 m breaststroke sequence yielded a mean absolute split-time error of 1.005 s and an RMSE of 1.210 s across 18 checkpoints, demonstrating the feasibility of metric-accurate swimmer tracking without specialised hardware. Wrist keypoint tracking remained noisy in the underwater environment, highlighting distal pose estimation as the primary target for future refinement.

Acknowledgements

Completing this project has been simultaneously one of the most humbling and rewarding experiences of my life, and it wouldn't have been possible without those who supported me along the way.

Much like a swimming coach, Dr. Vanessa Camilleri, you have been a guiding force throughout this project, providing me with the tools, feedback, and encouragement I needed to navigate the turbulent waters of research. I am deeply grateful for your mentorship, patience, and belief in my abilities, which have been integral to my growth as a researcher and success in this project.

Just as the water holds up a swimmer, my family has been my support system. Keeping me afloat during the long hours of work and study, your love and encouragement have been the lifeline that kept me going.

Similar to how the lanerope stays by a swimmer's side, my friends and peers have been consistently present throughout this journey, providing companionship, distractions, and much-needed breaks that kept me sane. The moments we've shared will always be cherished, and I am grateful for the laughter and support that helped me through the challenges of this project.

As a swimming rival pushes one to improve themselves, I would like to thank my lecturers for inspiring the curiosity and thirst for learning that drove me to pursue this project in the first place. Your dedication to teaching and research has been a constant source of motivation, and I am grateful for the education and opportunities you have provided me.

Finally, as I've run out of swimming metaphors, I would like to address the swimming community, whose dedication and love for the sport genuinely inspire me to push my own boundaries. Thank you for showing me what passion and commitment looks like, and what never giving up really means.

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1 Introduction

1.1 Problem Statement

Performance analysis in competitive swimming is traditionally reliant on manual observation or expensive, sensor-based systems that can be intrusive to the athlete [1] While computer vision offers a non-intrusive alternative, the aquatic environment introduces significant noise. Light refraction at the air-water interface, surface turbulence caused by the swimmer, and occlusions like bubbles and splashes degrade the quality of visual data [2, 3]. Furthermore, environment conditions, camera distances and angles vary significantly between venues, making consistent metric quantification, such as speed in meters/second, difficult without distinct calibration markers.

1.2 Motivation

There is a distinct lack of standardized, automated systems for extracting swimming metrics from video footage that do not require complex setups. A standard, simple, and well documented system capable of extracting metric data from video with minimal setup could act as a proof of concept, to prove the viability of such systems in aquatic environments. This project focuses on validating feasibility and robustness rather than achieving state-of-the-art performance.

1.3 Aims and Objectives

The primary aim of this project is to develop a Proof of Concept system that enables the extraction of swimming metrics from video footage. The specific objectives are:

1. To design, prototype, and create a modular, under-water marker design that maximizes underwater visibility.
2. To create and annotate a small dataset of swimming footage using the designed markers in the background.
3. To establish a pipeline for dynamic pixel-to-meter scale extraction using the markers for calibration.
4. To utilize the calibrated scale to provide useful kinematic metrics such as swimmer velocity and trajectory.

1.4 Project Structure

The remainder of this report is structured as follows.

Chapter 2– Background This chapter provides an overview of the core concepts and technological advancements relevant to the project.

Chapter 3– Literature Review This chapter reviews existing research and methodologies in the field of computer vision for sports analytics, with a focus on swimming performance analysis. It identifies gaps in current approaches and situates the proposed system within the broader context of the use of AI in sports technology.

Chapter 4– Methodology This chapter details the proposed methodology for developing the proof-of-concept system, including the design of underwater markers, dataset creation, and the calibration pipeline for dynamic scale extraction.

Chapter 5– Evaluation This chapter presents the evaluation of the proposed system, including assessments of each step of the pipeline, and an overall assessment of the system's performance in extracting swimming metrics from video footage.

Chapter 6– Conclusion and Future Work This chapter summarizes the findings of the project, discusses the implications of the results, and outlines potential directions for future research and development in this area.

2 Background

Existing literature in sports biomechanics highlights the importance of kinematic metrics like acceleration and hand trajectory. While computer vision research has successfully applied deep learning for detection, classification, segmentation, and human pose estimation (HPE), a critical limitation in many CV-based swimming systems is the conversion of distances from the pixel space to real-world units, often relying on relative metrics or requiring external sensors.

This chapter covers the relevant foundations upon which the proposed system is built, including object detection and classification, semantic segmentation, and human pose estimation. The chapter concludes with a brief summary of the discussed techniques.

2.1 Object Detection and Classification

Object detection is a fundamental computer vision task that involves identifying the presence and location of objects within an image. In the context of swimming analysis, this typically involves localizing the swimmer and specific environmental markers. Modern detectors have shifted from region-based convolutional neural networks (R-CNNs) to "one-stage" detectors to optimize for speed and real-time performance.

The YOLO (You Only Look Once) series represents the state-of-the-art in one-stage detection. The iterations used in this work, YOLOv8 and YOLO26 [4], introduces an anchor-free detection mechanism. Unlike previous versions that relied on predefined anchor boxes, anchor-free models predict the centre of an object directly, which reduces the complexity of the non-maximum suppression (NMS) stage. This results in improved performance on varying scales, which is a critical feature when tracking markers that change size due to camera distance or water refraction. In sports analytics, YOLO-based architectures are favoured for their ability to process high-frame-rate video without significant latency [5].

2.2 Semantic Segmentation

Semantic segmentation involves assigning a categorical label to every pixel in an image, facilitating a more granular understanding of the scene than bounding boxes. For aquatic environments, segmentation is used to isolate the swimmer's body from highly textured backgrounds or to precisely delineate markers amidst bubbles and surface turbulence.

A significant advancement in this field is the Segment Anything Model (SAM) [6]. SAM is a "foundation model" trained on over a billion masks, enabling zero-shot generalization to new domains, such as underwater footage, without additional training. This project utilizes SAM 2 [7], which extends this capability to the temporal domain through a memory-based attention mechanism. SAM 2 treats video as a continuous stream, using a “memory bank” to store features from previous frames. This allows the system to maintain a stable segmentation of markers even during transient occlusions or periods of significant water splash, addressing a core problem identified in the aquatic environment.

In addition to its role in tracking, SAM 2 was utilized in this research as a semi-automated annotation tool to facilitate dataset creation. Because manual pixel-level annotation of video sequences is prohibitively time-consuming, SAM 2’s temporal propagation feature was used to generate high-fidelity masks for underwater markers across several video sequences from a few manual prompts. These generated masks were then converted into bounding boxes and used to fine-tune a custom YOLO26 model for marker detection. This “model-in-the-loop” approach enables the creation of robust, domain-specific detectors with significantly reduced manual labelling effort. It bridges the gap between general-purpose foundation models and specialized sports analytics tasks.

2.3 Human Pose Estimation

Human Pose Estimation (HPE) is the process of identifying and tracking the coordinates of specific anatomical keypoints, such as joints and extremities. In swimming, HPE is used to calculate stroke frequency and joint angles. Traditionally, HPE utilized High-Resolution Nets (HRNet) to maintain high-resolution representations throughout the network.

However, recent research has transitioned toward Transformer-based architectures. ViTPose [8] utilizes a non-hierarchical Vision Transformer (ViT) [9] as its backbone. Unlike convolutional neural networks (CNNs) that process images through local receptive fields, Transformers use self-attention to capture long-range dependencies across the entire frame. This architectural choice is vital for the aquatic domain. While CNNs are limited by local receptive fields, the global self-attention in ViTPose allows the model to leverage the context of the swimmer’s torso to infer the position of distal keypoints—like the wrist or ankle—even when they are visually obliterated by surface splashes or lost in high-frequency bubbles. This moves beyond simple detection toward a spatial reasoning approach. This project utilizes the large-scale variant of ViTPose to ensure high precision in keypoint localization before converting these coordinates into metric data through marker-based linear scaling and baseline projection.

2.4 Motion Compensation and Ego-motion

In sports videography, unless the camera is rigidly fixed to a stationary mount, the resulting footage contains “ego-motion”, motion caused by the movement of the camera itself. For swimming analysis, where a coach may follow the athlete along the poolside, this introduces a significant challenge: the pixel coordinates of the swimmer are a summation of the swimmer’s actual velocity and the camera’s displacement.

To resolve this, computer vision systems utilize stationary environmental features, such as lane ropes or pool-floor markings, as a global coordinate system. By tracking these fixed points using a homography matrix or simpler translation vectors [10], the system can perform background registration. This process effectively “stabilizes” the coordinate space, allowing for the extraction of the swimmer’s true kinematic trajectory relative to the pool rather than the image frame.

2.5 State Estimation and Trajectory Smoothing

Raw data extracted from deep learning models often contains high-frequency noise or “jitter” caused by partial occlusions, light refraction, or fluctuating confidence scores in the model’s output. To transform these noisy observations into a physically plausible trajectory, state estimation techniques are employed.

The Kalman Filter [11] is the standard recursive algorithm for this purpose. It operates in two phases: a “predict” step, which estimates the current state (position and velocity) based on previous data and Newtonian laws of motion, and an “update” step, which adjusts this prediction based on the new noisy measurement from the CV model. This approach is particularly effective in swimming because it can bridge gaps in detection, such as when a swimmer’s limb goes out of view or becomes obscured by a splash, by relying on the predicted momentum of the limb until it reappears.

2.6 Summary

The integration of YOLO for rapid detection, SAM 2 for robust temporal segmentation and automated dataset generation, and ViTPose for high-fidelity pose estimation forms a comprehensive, modular pipeline for swimming analysis. These tools collectively address the core challenges of the aquatic environment—namely visual noise, light refraction, and the historical lack of labelled underwater data—by leveraging recent breakthroughs in anchor-free detection, memory-based attention, and global spatial transformers.

3 Literature Review

3.1 Evolution of AI in Sports Analytics

Computer vision (CV) has fundamentally altered the landscape of sports analytics, transitioning from manual annotation to automated, high-fidelity measurement. Tracing the historical arc of this evolution reveals a trajectory that began in the 1990s and early 2000s with manual notational analysis. During this era, practitioners relied on painstaking frame-by-frame video review and manual timing systems to derive basic performance metrics. The subsequent era of “hand-crafted” features saw the introduction of algorithms such as the Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and various optical flow techniques. While these early systems represented a step toward automation, they relied on heuristic-based observation and were notoriously fragile, frequently failing under the non-uniform lighting and high-entropy background noise characteristic of aquatic environments.

The paradigm shift occurred following the “Deep Learning revolution”, catalysed in large part by the success of AlexNet in the 2012 ImageNet challenge [12]. This milestone demonstrated that hierarchical feature extraction through Deep Convolutional Neural Networks (CNNs) could outperform any human-engineered descriptor. Building on the foundations laid by earlier architectures such as LeNet [13], researchers began applying deep learning to sports-specific tasks. In the mid-2010s, models such as ResNet [14] became standard for robust feature representation, allowing for more stable tracking in dynamic environments.

However, the adoption of these technologies has been uneven across the sporting landscape. Field sports such as football, tennis, and athletics have received a disproportionate share of research attention due to the relative ease of data collection in well-lit, terrestrial environments. In contrast, swimming has lagged behind, primarily due to the unique optical challenges of the pool. As noted by [15], AI-driven systems are now the standard for extracting kinematic data and estimating pose in real time, but their application in swimming remains a frontier of computer vision. The current paradigm is moving toward Foundation Models, defined as large-scale models trained on massive, diverse datasets that exhibit zero-shot generalization [9]. This thesis utilizes models such as SAM 2 and ViTPose, which provide a level of geometric and spatial reasoning, as well as temporal consistency, previously unavailable in sports-specific contexts. Tools such as DeepLabCut [16] and OpenPose [17] pioneered the transfer of these capabilities to biomechanics, but the transition to the aquatic domain requires a fundamental reassessment of how models perceive motion through water.

3.2 An Adversarial Domain for Computer Vision

While field sports such as football or athletics provide relatively stable visual environments with predictable occlusions, the aquatic setting serves as an adversarial domain that stress-tests the limits of image processing. Unlike terrestrial pose estimation, where the medium of air is the standard, the water-air interface introduces non-linear geometric distortions that violate the fundamental pinhole camera model assumptions [18].

To understand the severity of this challenge, one must consider Snell's Law, which dictates that $n_1 \sin\theta_1 = n_2 \sin\theta_2$. Because the refractive index of water ($n \approx 1.33$) differs significantly from air ($n \approx 1.00$), the apparent position of a swimmer's joint underwater is displaced from its true anatomical coordinate [19]. Underwater camera systems require careful calibration to compensate for this distortion, often necessitating specialized refractive camera models or dedicated underwater calibration rigs [20]. Without such compensation, standard pose estimation models produce systematic errors in joint angle calculations, rendering them insufficient for high-performance biomechanical analysis.

Beyond refraction, the environment is plagued by dynamic noise. As [3] highlights, measurements are strongly influenced by disruptive elements such as bubbles, splashes, and surface glare. These elements introduce severe, transient occlusions and "caustics", the dancing light patterns on the pool floor caused by the focusing effect of surface waves. Indoor pool lighting often exacerbates these effects, creating high-contrast reflections that can be mistaken for anatomical features by naive detectors. Furthermore, swimming involves a unique multiphase appearance problem: an athlete alternates between above-water and below-water states. During a butterfly stroke recovery or a freestyle tumble turn, the swimmer's body undergoes sudden, drastic changes in visual texture and limb visibility that have no direct terrestrial analogue.

These environmental stressors explain why terrestrial benchmarks, such as COCO [21] or the MPII Human Pose dataset [22], are largely inadequate for the aquatic domain. While a model trained on COCO may excel at identifying a pedestrian on a pavement, it lacks exposure to the blurred, refracted, and partially occluded silhouettes of a submerged athlete. Einfalt et al. [23] demonstrate that even fine-tuned CNNs evaluated on real swimming footage exhibit systematic error modes in the aquatic environment, confirming that models pre-trained on terrestrial data generalize poorly to this domain. This necessitates the use of temporal memory mechanisms, such as the memory bank implemented in SAM 2 [7], which allows a system to maintain object features even when they are obscured by a splash or submerged beyond clear visibility. By treating the aquatic environment as a structured noise problem rather than random interference, temporal dependencies can be exploited to maintain tracking where classical optical flow methods fail systematically.

3.3 From Handcrafted Features to Foundation Models

Historically, structural feature detection in swimming relied on the Hough Transform or colour-keying to isolate lane ropes or markers. However, these methods break down under varying pool illumination and water turbulence [24]. Colour-keying, for instance, is highly susceptible to the "blue-shift" of underwater light, whereby longer wavelengths are absorbed more rapidly, causing markers to lose their distinct chromatic signatures at depth. Similarly, classical optical flow algorithms such as Lucas-Kanade fail in the presence of "aperture problems" caused by the repetitive, textureless nature of pool tiles and the erratic motion of bubbles.

The subsequent shift toward CNNs, including U-Net and earlier You Only Look Once (YOLO) variants, offered better robustness to noise. In the context of pose estimation, this era saw the emergence of two primary strategies: "bottom-up" and "top-down" approaches. Bottom-up models such as OpenPose [17] detect all body parts in an image and then group them into individuals, which is computationally efficient but often struggles with the limb-assignment ambiguities found in a crowded swimming lane. Top-down approaches first detect the person and then perform pose estimation within that bounding box. For instance, [25] demonstrated the utility of YOLOv8 for athlete identification, providing a stable crop for subsequent analysis. Nevertheless, CNNs are often limited by their local receptive fields. Because a convolution operation only 'sees' a small neighbourhood of pixels at a time, it can struggle with the long-range spatial dependencies required to identify a swimmer's full limb extension when the torso is clear but the hand is obscured by a splash.

These limitations led to the adoption of Vision Transformers (ViT). Unlike convolutions, the self-attention mechanism in ViT [9] allows every token in the frame to interact with every other token. Models such as ViTPose [8] exploit this global context to process the entire frame simultaneously, allowing the model to leverage the context of the swimmer's torso to accurately predict the position of a wrist or ankle, even if that specific joint is visually distorted by surface ripples. ViTPose is particularly notable for its "plain ViT" backbone and simple decoder head, an architectural choice that prioritizes feature richness over complex structural engineering and tends to improve robustness on out-of-distribution data such as underwater footage.

The most recent evolution is the emergence of Foundation Models for segmentation and feature extraction, notably the Segment Anything Model (SAM) [6] and DINO [26]. SAM 2 extends segmentation to the temporal domain through a dedicated memory bank, enabling the segmentation of arbitrary objects such as underwater calibration markers with zero-shot accuracy. These models offer strong geometric reasoning, though not without trade-offs: the computational cost of global self-attention is considerably higher than that of local convolutions, which presents challenges for real-time deployment. For high-fidelity biomechanical research, however, where accuracy takes precedence over latency, the capacity of Transformers to maintain a coherent kinematic chain through a turbulent water-air interface represents a meaningful advance over the CNN era.

3.4 The Challenge of Metric Scaling and Calibration

A fundamental limitation in many CV-based swimming systems is the conversion of distance from pixel space to real-world, metric units. A velocity of x pixels per second is meaningless without a reference scale. In sports biomechanics, the "gold standard" for this conversion is 3D multi-camera calibration, often using Direct Linear Transformation (DLT). As the constraints of multi-camera acquisition in swimming demonstrate [27], poolside data collection is limited by installation space, the requirement for waterproof housing, and the difficulty of camera synchronization, making this approach impractical in routine coaching environments.

Alternative paradigms include wearable Inertial Measurement Units (IMUs), which can provide direct acceleration data. However, IMUs face significant challenges in swimming, including data contamination from water resistance and the fact that they are often prohibited in official competitions. This leaves CV-based planar homography as the most viable alternative. Homography assumes a mapping between two planes, for example the plane of the swimmer's lane and the camera sensor. While [28] utilized lane ropes for this purpose, discrete, underwater markers provide a more stable reference.

A critical issue in homography-based scaling is error propagation. Because velocity is a derived quantity ($v = \Delta d / \Delta t$), a small error in marker detection or a slight violation of the "planar world" assumption (for example, the swimmer moving deeper into the pool than the calibration plane) can lead to significant inaccuracies in stroke length and speed calculations. This project extends the logic of marker-based calibration by using SAM 2 to track these markers with high temporal consistency. By establishing a fixed pixel-to-metre ratio based on known marker distances, such as the 2.5 m proposed in this thesis, the system achieves metric accuracy without the overhead of an expensive sensor array or the instability of manual scaling.

3.5 Addressing the Data Scarcity Gap: Model-in-the-Loop Annotation

A persistent gap identified in the literature is the scarcity of publicly available, annotated underwater datasets. To quantify this gap, one need only compare the scale of terrestrial datasets such as COCO, which contains over 200,000 labelled images [21], with aquatic datasets such as SwimmerNET [3], which are orders of magnitude smaller and often lack specific features such as underwater markers. This scarcity is driven by ethical restrictions on filming athletes, the logistical difficulty of underwater data collection, and a lack of standardized annotation protocols for the aquatic domain.

A solution emerging in recent literature is Model-in-the-Loop (MITL) annotation, which sits within the broader landscape of semi-supervised learning. Traditional semi-supervised techniques such as pseudo-labelling [29] and consistency regularization (for example, FixMatch [30]) attempt to leverage unlabelled data by having a model predict its own labels. MITL refines this by using high-capacity Foundation Models to generate pseudo-labels. For example, by using SAM 2's zero-shot segmentation to identify markers in a small subset of frames and propagating those masks through the temporal sequence, a high-quality training set can be generated for a more efficient, task-specific detector such as YOLOv8.

This approach is further supported by self-supervised models such as DINOv2 [31], which learn robust visual features without any labels at all. However, MITL carries risks, the primary concern being "confirmation bias", whereby the student model (for example, YOLO) learns to replicate the systematic errors of the teacher model (for example, SAM 2). This project mitigates such risks through human-in-the-loop verification, ensuring that the automated MITL pipeline produces markers that are anatomically and geometrically sound. This workflow represents a departure from traditional manual annotation and directly addresses the research need for domain-specific underwater features.

3.6 Summary of Research Needs

The reviewed literature confirms a critical tension between the high-performance requirements of swimming biomechanics and the adversarial nature of the aquatic environment. The transition from manual notation to deep learning has provided the tools for automation, yet the field remains hampered by three persistent challenges: the failure of traditional trackers during high-turbulence occlusions; metric inaccuracy caused by refraction and poor calibration; and the "cold start" problem resulting from a lack of labelled underwater data.

This research bridges these gaps by synthesizing a multi-stage pipeline that leverages recent advances in Foundation Models. By using SAM 2 for stable temporal segmentation, the challenge of occlusion and splash noise is addressed directly. By implementing a marker-based homography, a robust solution to the metric scaling problem is provided that accounts for the refractive properties of the medium. Finally, by employing an automated MITL pipeline, the work demonstrates a scalable method for generating domain-specific data without the prohibitive cost of solely manual annotation. Ultimately, this work moves beyond qualitative video analysis toward a metric-accurate system that brings the rigour of terrestrial sports analytics to the complex, fluid environment of competitive swimming.

4 Methodology

This chapter details the technical implementation of the proof-of-concept system. The methodology is structured around the four primary project objectives, transitioning from physical marker design to the final kinematic extraction engine. Critically, each phase was designed not in isolation but in direct response to the failure modes introduced by the preceding phase: the marker design constrains what the detector must handle, the detector's output feeds the scale calibration, and the calibration underpins the accuracy of every kinematic metric. The decisions documented here are therefore presented as a chain of engineering arguments rather than a sequence of independent implementation choices.

4.1 Marker Design and Visibility

The first objective required a modular marker design capable of withstanding the high-turbulence environment of a swimming pool while remaining identifiable under varying refractive conditions.

The lack of existing documentation on underwater marker design for computer vision applications necessitated an iterative prototyping approach. This led to the decision to create a modular design for a 3D-printed marker that could be easily set up along the lane. The final design, as shown in Figure 4.1, consists of high-contrast, alternating black and white segments, which seamlessly screw into one another. This decision to use a modular design allowed us to test various configurations, mainly how many segments to stack, effectively altering the height of the marker.

4.1.1 Pattern Selection: High-Contrast Binary Alternation

The choice of an alternating black-and-white pattern over coloured or single-colour designs was motivated by the specific optical failure modes of the aquatic environment. Coloured markers suffer from wavelength-dependent attenuation: water selectively absorbs red and orange wavelengths within the first few metres of depth, causing a progressive blue-shift that desaturates warm tones and renders chromatic patterns highly variable across different pool depths and lighting conditions. Fluorescent pool lighting compounds this problem by introducing a spectrally narrow illuminant that can cause metameric failure, where colours that appear distinct under natural light become perceptually indistinguishable under artificial spectra. Monochromatic high-contrast patterns sidestep these failure modes entirely because detection relies on luminance gradients rather than hue. A binary pattern of maximum contrast, such as pure black against

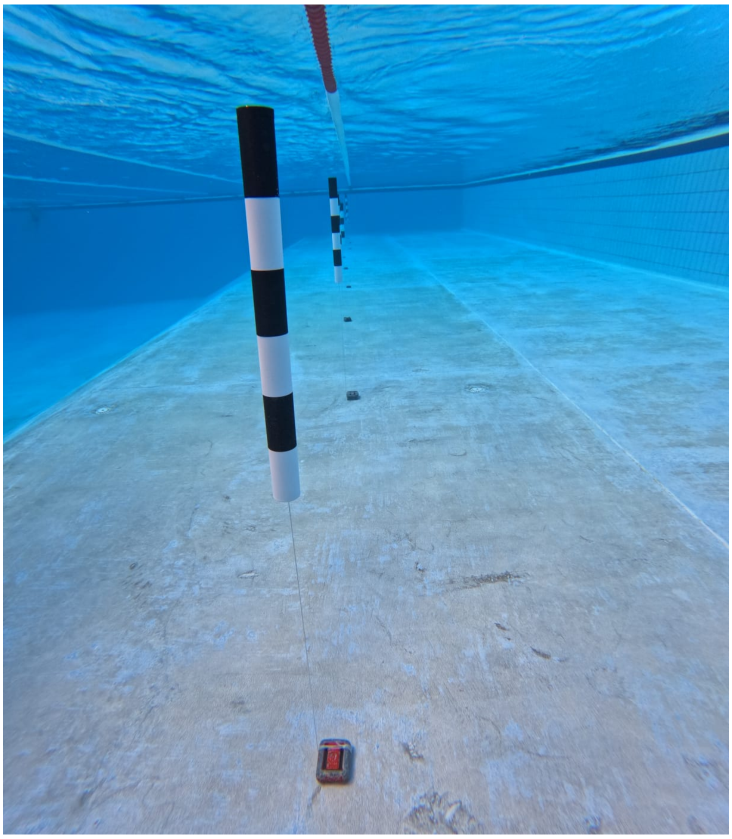

Figure 4.1 An image of the final marker design in use.

pure white, retains a large gradient even after the global contrast reduction caused by water's scattering coefficient and the distorting effects of refraction at the air-water interface [18].

The alternating structure, rather than a solid single-colour cylinder, was chosen for two further reasons. First, a solid black cylinder placed near a dark pool floor or against a dark swimsuit provides insufficient contrast for reliable edge detection, whereas an alternating pattern creates multiple distinct transitions detectable by gradient-based feature extraction regardless of the local background. Second, the periodic alternation constitutes a unique visual signature that distinguishes the marker from other cylindrical objects present in the scene, most notably lane rope buoys, which are typically uniform in colour along their length. This distinction is exploited implicitly by the downstream detector, which learns that the alternating pattern is a necessary characteristic of a valid detection.

4.1.2 Physical Form Factor: Cylindrical and 3D-Printed

Off-the-shelf calibration targets such as chequerboard boards or ArUco markers were considered as alternatives. These planar targets are well-characterized in the literature [10] and are the standard choice for metric calibration in controlled laboratory settings. However, they present a fundamental geometric limitation in the poolside context: their detectability is highly sensitive to viewing angle. At the oblique angles typical of handheld poolside footage, a flat board subtends a narrow projected area and its corner features become unreliable, degrading calibration accuracy precisely in the conditions under which the system must operate. No commercially available waterproof, poolside-mountable calibration target of appropriate scale was identified during the design phase, reinforcing the case for a bespoke solution.

A cylindrical form was selected because rotational symmetry about the vertical axis guarantees that the marker presents a geometrically consistent cross-section from any horizontal viewing direction. Whether the operator films from directly to the side or at a moderate angle, the apparent width of the cylinder changes only by the cosine of the viewing angle, and within the moderate angular range achievable from a poolside position, this variation is small. The decision to 3D-print the markers allowed rapid iteration over design parameters and ensured the final geometry was precisely controlled.

Each segment is 10 cm in height, with a 5 cm diameter, and the alternating pattern provides a unique visual signature that can be reliably detected by computer vision algorithms. The design also incorporates a screw mechanism that allows for easy assembly and disassembly, enabling quick setup along the lane for calibration purposes. This modularity not only facilitates testing different configurations but also ensures that the markers can be easily transported and stored when not in use. The 10 cm segment height was determined by the requirement that the marker subtend a sufficient number of pixels to be reliably detected at the maximum anticipated operating distance: at a camera-to-marker distance of approximately two to three metres, a 10 cm feature projects to roughly 50 pixels at a 1080p resolution, which is sufficient for a YOLO-family one-stage detector to localize accurately. A smaller segment height would have reduced this pixel count below the threshold of reliable detection, while a larger height would have increased physical mass and the risk of the marker toppling in turbulent conditions.

4.1.3 Marker Spacing and Empirical Calibration

The spacing between the markers was determined based on multiple tests, which revealed that an interval of 2.5 m provided a good balance between visibility and the number of markers required for accurate calibration. This spacing allows for sufficient coverage along the lane while minimizing the number of markers needed, which is crucial for minimizing setup time and complexity.

The spacing decision involved explicit consideration of the failure modes at both extremes. A 1 m spacing would increase the total number of markers required to cover a 25 m or 50 m lane, substantially raising setup complexity and the likelihood of inter-marker occlusion as the swimmer's body passes between the camera and multiple adjacent markers simultaneously. A 5 m spacing would reduce the probability of two or more markers being simultaneously visible within the camera's field of view at any given moment, which is the minimum condition for a stable pixel-to-metre ratio estimate. With only a single marker visible, no inter-marker distance can be computed, and the scale calibration must fall back to a carried-forward estimate from the previous valid frame, introducing temporal staleness. The 2.5 m spacing was identified through empirical trials as the minimum spacing at which at least two markers reliably remained within the camera's field of view throughout the swimmer's traversal of the monitored segment, given the focal length and field of view of the camera used in the trials. This provides a stable baseline for pixel-to-metre ratio estimation at every frame, minimizing reliance on the carry-forward fallback described in Section 4.3.

4.1.4 Prototyping Iterations

Early prototypes explored solid single-colour cylinders, different diameters, and varying numbers of stacked segments. Solid-colour prototypes exhibited the contrast failure modes described above under pool conditions, particularly when the camera position placed the marker against a similarly toned background. Markers with a diameter smaller than 5 cm proved difficult to detect reliably beyond 2 m, while larger diameters increased buoyancy and required heavier weighting to remain upright. These observations collectively converged on the final configuration: a 5 cm diameter cylinder with 10 cm alternating segments, spaced at 2.5 m intervals.

4.2 Dataset Generation and Model-in-the-Loop Annotation

To address the second objective, necessitated by the lack of annotated underwater datasets, a "Model-in-the-Loop" (MITL) workflow was developed. This phase utilized foundation models to automate the labelling process for the downstream YOLO26 detector. The entire dataset creation pipeline is visualized in Figure 4.2.

4.2.1 Motivation for the MITL Approach

Before describing the implementation, it is necessary to justify why the MITL paradigm was selected over the three principal alternatives. Full manual annotation was the most obvious baseline: a trained annotator can produce a bounding box label in approximately two to five minutes per frame when working carefully in visually challenging material. At 25 frames per second, even a single 60-second clip yields 1,500 frames. The annotation cost of thousands of frames is therefore completely intractable as a single-phase activity, particularly given the iterative nature of the dataset quality refinement described in the

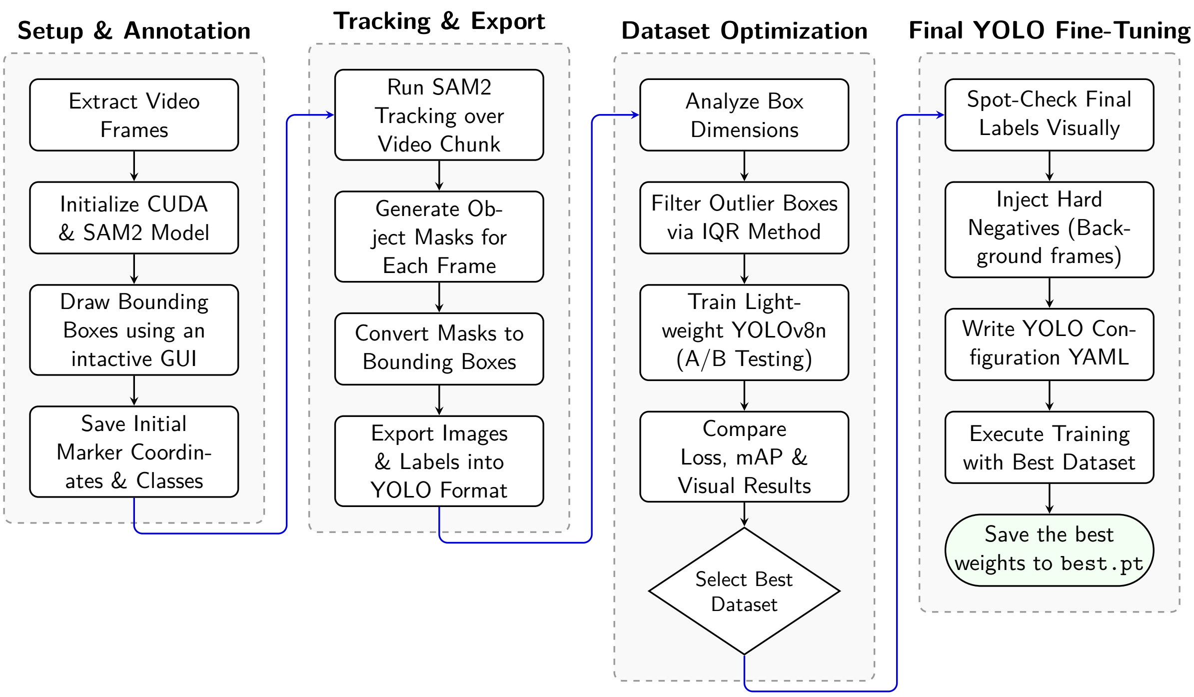

Figure 4.2 Marker Detection Model Training Pipeline.

following subsections. Synthetic data generation was evaluated as a potential alternative but was rejected because faithful simulation of real underwater optical phenomena, including caustic light patterns, refractive distortion at the air-water interface, and the turbidity-dependent scattering of pool water, remains an unsolved problem in rendering. A domain gap of this magnitude between synthetic and real data would require explicit domain adaptation, introducing a layer of complexity disproportionate to the single-class detection task. Transfer learning from terrestrial calibration target datasets was similarly infeasible: no public dataset of annotated underwater cylindrical calibration targets exists, and the visual characteristics of the markers, strongly shaped by the aquatic optical environment, are sufficiently different from terrestrial equivalents to make direct transfer learning unreliable without substantial domain adaptation. The MITL approach, by contrast, exploited the zero-shot generalization capability of a large foundation model to generate approximate labels at near-zero marginal cost, with subsequent stages refining quality to the standard required for training a precision detector.

4.2.2 Automated Labelling with SAM 2

The Segment Anything Model 2 (SAM 2) [7] was utilized to propagate marker bounding boxes across video sequences. Prompts were provided from the initial frames, propagated through the video. Once all tracked markers exited the frame, the process was repeated for the next chunk of frames until the entire video was annotated. This approach reduced manual annotation time substantially while maintaining high-quality labels. This process was repeated for multiple videos, resulting in a dataset of over 4,000 annotated frames, and over 7,000 bounding boxes.

SAM 2 was selected over the original SAM v1 because its architecture incorporates a temporal memory bank that allows object identity to be propagated across video frames rather than requiring independent prompting of each image [7]. For a video annotation task, this temporal propagation is the core requirement: without it, the model is simply applied frame-by-frame with no continuity guarantee, offering no advantage over per-frame automated segmentation followed by bounding box extraction. The memory bank design means that an object prompted in frame t can be tracked through subsequent frames without further human input, which is precisely the efficiency gain the MITL pipeline depends on.

Propagation was performed in chunks rather than as a single pass over each video for a principled reason. Since the video footage follows the swimmer along the lane, markers will be popping in and out of the frame as the camera pans. If the entire video is processed in one pass, once all the tracked objects are out of the frame, the model will have no active prompts to track and will effectively be operating in zero-shot mode, producing unreliable segmentations. By processing in chunks, the model is reset with new prompts at regular intervals, ensuring that it always has a valid reference for tracking and preventing the degradation of segmentation quality over time. Before ingestion into SAM 2, frames were extracted from video at the native frame rate to preserve temporal continuity, and no resolution downsampling was applied, as the markers' small pixel footprint made preserving full resolution important for accurate mask boundaries.

4.2.3 Outlier Filtering

Automated tracking via foundation models, while efficient, is susceptible to "mask jitter" and drift, particularly when visual features are distorted by water refraction or transient occlusions. To ensure the robustness of the downstream YOLO detector, a statistical filtering stage was implemented to prune the raw SAM 2 output.

The filtering process focuses on the geometric consistency of the generated bounding boxes. Based on the assumption that the physical dimensions of the markers are constant, any significant variation in the predicted box area or aspect ratio was flagged as tracking noise. The system utilizes the Interquartile Range (IQR) method [32] to identify and remove these outliers. For a set of bounding box areas A, the filtering bounds are defined as:

$$[\text{Lower Bound}, \text{Upper Bound}] = [Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}] \tag{4.1}$$

where Q_1 and Q_3 represent the first and third quartiles, respectively.

The IQR was preferred over the standard deviation as the basis for outlier detection because the standard deviation is a non-robust statistic: it is sensitive to the very extreme values it is attempting to identify, since each outlier inflates both the mean and the standard deviation and thereby raises the rejection threshold [33]. The IQR, by contrast, is computed entirely from the central half of the distribution and is therefore resistant to extreme values by construction. The Tukey fence multiplier of 1.5 was retained rather than adopting a stricter value such as 3.0, because the goal at this stage is to produce a clean training set rather than to preserve maximum data volume: a false negative (retaining a corrupted label) is more damaging to downstream model quality than a false positive (discarding a valid label), and the 1.5 multiplier provides a conservative rejection criterion appropriate for this asymmetric cost.

The specific failure modes observed in the raw SAM 2 output fell into two categories. The first was mask bleed, wherein SAM 2 expanded a marker's mask boundary to incorporate an adjacent splash event, substantially increasing the bounding box area and reducing the inferred aspect ratio. The second was identity switching, wherein the tracker attached to a lane rope buoy rather than the intended marker, producing a bounding box of markedly different aspect ratio. Filtering on both area and aspect ratio jointly was therefore necessary: area filtering alone would not have caught identity switches if the buoy happened to subtend a similar projected area to the marker, while aspect ratio filtering alone would not have caught mask bleed events that preserved the elongated vertical shape of the marker while expanding its width. Any frame containing boxes falling outside these bounds was excluded from the training set. This ensured that the model was trained only on high-confidence, geometrically plausible representations of the underwater markers.

4.2.4 Dataset Selection

To ensure the YOLO26 model was trained using the best possible dataset, an A/B testing approach was implemented. Two datasets were created: Dataset A, which included all SAM-generated labels, and Dataset B, which included only the filtered labels after the IQR outlier removal process. Both datasets were used to train separate YOLO26n models, and their performance was compared using standard metrics such as mean Average Precision (mAP) and visual inspection of the predicted bounding boxes.

The nano variant was used for the comparative experiment rather than the small variant that was ultimately deployed, because the lower training time of the nano architecture allows the A/B comparison to be completed within a practical iteration cycle. If the quality benefit of filtering is genuine, it will manifest at nano scale as a consistent improvement in mAP across evaluation thresholds, constituting sufficient evidence to justify applying the same pipeline to the larger model. The filtered dataset yielded a consistently higher mAP across all evaluation thresholds, confirming that the IQR removal stage improved training data quality in a measurable way, thus was selected for the final training of the YOLO26s model.

4.2.5 Human-in-the-Loop Verification

To ensure the high precision required for metric extraction, a custom-developed graphical verification tool was utilized to perform a final audit of the filtered dataset. This tool, shown in Figure 4.3, implemented using the Tkinter framework, allowed for the rapid visual inspection of the SAM-generated labels.

The interface displayed all the video frames with their bounding boxes overlaid, allowing the user to quickly navigate through the entire dataset. The user could click on any frame to maximize it, and then manually add, remove, or adjust bounding boxes as needed. This step was crucial for catching any remaining errors that automated filtering might have missed, such as subtle mask bleed or borderline identity switches that did not trigger the IQR outlier detection. Thanks to the initial filtering, the main use of this tool was to manually add bounding boxes for frames where SAM 2 failed to detect a marker at all, which is a critical failure mode for scale estimation since it reduces the number of visible reference pairs. The tool's design prioritized efficiency, allowing the user to audit thousands of frames in a matter of hours, thus ensuring that the final training set was of the highest possible quality before being used to train the YOLO26s model. This hybrid approach of combining foundation model automation with human oversight exemplifies the MITL paradigm, leveraging the strengths of both machine and human intelligence to create a robust dataset for training a precision detection model. This step was crucial for establishing a ground truth used for the final model fine-tuning and the subsequent accuracy evaluations in Chapter 5.

4.3 Dynamic Scale Extraction Pipeline

Objective 3 focuses on establishing a stable reference for converting image coordinates into metric units.

4.3.1 Model Selection Rationale

Before describing the training configuration, it is necessary to justify the choice of YOLO26 over alternative detection architectures. Two-stage detectors such as Faster R-CNN achieve higher accuracy on standard benchmarks by separating region proposal from classification, but their inference speed is incompatible with the requirement to process

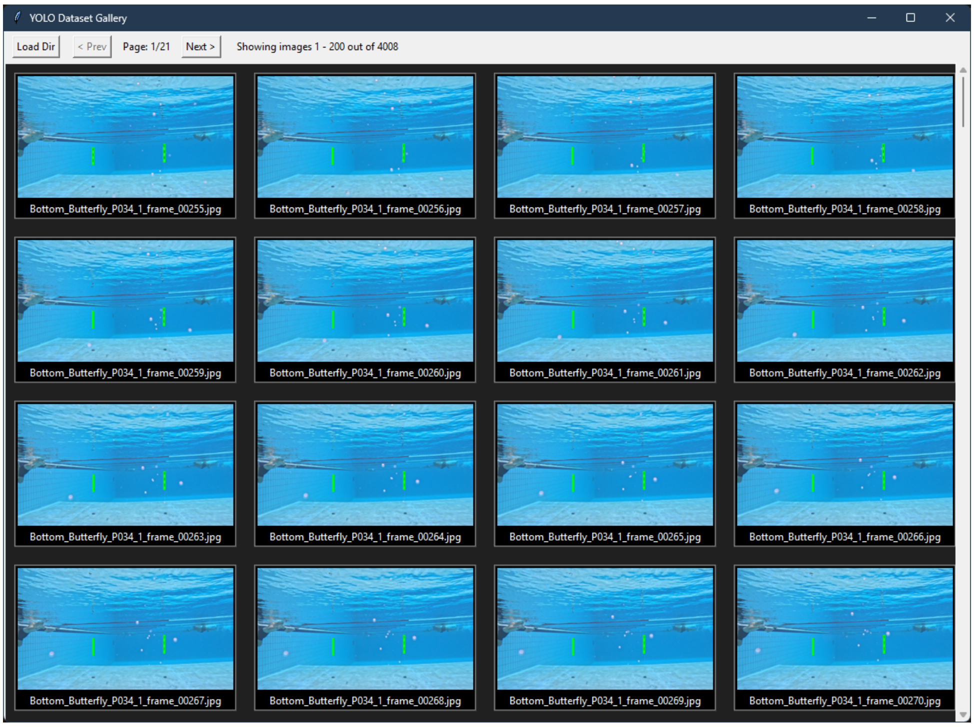

Figure 4.3 Human-in-the-loop verification tool interface for annotated bounding boxes.

full video sequences at interactive rates. For a single-class detection task on visually distinctive markers, a one-stage YOLO variant provides sufficient localization accuracy at a fraction of the computational cost, making the accuracy-throughput trade-off clearly favourable for the proposed pipeline. Among the YOLO family, earlier versions such as v7, v8, and v10 were considered but YOLO26 was selected for its improved small-object detection head and its anchor-free regression formulation, which is better suited to cylindrical targets whose bounding box aspect ratios vary continuously with viewing distance rather than clustering around a few predefined anchor aspect ratios.

4.3.2 Marker Detection via YOLO26

A custom YOLO26s model [4] was fine-tuned on the SAM-generated dataset to provide a robust method for localizing markers. To ensure high localization accuracy, which is critical for the subsequent pixel-to-metre ratio calculation, the training process utilized a Distribution Focal Loss (DFL) gain of 1.5 and a Box Loss gain of 7.5. These settings prioritize bounding box regression over classification accuracy, as the project involves a single-class detection task where classification loss is trivially small and allocating representational capacity to it would be wasteful: upweighting the box regression terms forces the model to refine its localization precision, which is the property that directly governs the accuracy of the scale calibration downstream.

Training was conducted over 200 epochs with an image resolution of 640×640 pixels. Higher resolutions were considered but rejected on the grounds that the markers occupy sufficient pixels at 640 that sub-pixel localization precision is not the limiting factor in scale estimation, and the associated increase in GPU memory consumption would have required reducing the batch size, adversely affecting gradient estimate quality. To mitigate overfitting on the SAM-generated dataset, several regularization and augmentation strategies were employed.

An initial learning rate of 0.01 was used with a 3-epoch warm-up period and a momentum of 0.937 to stabilize early gradient descent, preventing the large gradient steps in the initial epochs from perturbing the pre-trained feature representations before the learning rate scheduler had converged. The model utilized Mosaic augmentation (1.0) and Random Erasing (0.4) to simulate occlusions and varying marker densities, forcing the network to learn robust features rather than memorizing specific marker placements. Mosaic augmentation directly addresses the scenario where markers appear at different apparent scales within the same field of view, as occurs when the camera is closer to some markers than others along the lane. Random Erasing simulates transient occlusions caused by splashing water, training the model to produce valid detections even when part of the marker is obscured. Variations in lighting and glare were accounted for by adjusting HSV (Hue, Saturation, Value) gains ($hsv\_h = 0.015$, $hsv\_s = 0.7$, $hsv\_v = 0.4$). The saturation jitter value of 0.7 is intentionally high because pool environments vary dramatically in colour temperature between outdoor venues under natural daylight and indoor venues under fluorescent or LED illumination, and the model must remain effective across this range without retraining. An early stopping patience of 20 epochs was implemented to terminate training if the validation fitness failed to improve, providing an automatic mechanism for balancing the risk of underfitting at low epoch counts against the risk of overfitting at high epoch counts given the limited dataset size. Training was performed on a CUDA-capable GPU, and the resulting model outputs high-confidence bounding boxes that serve as the foundation for physical distance estimation, leveraging the known dimensions of the physical markers to establish spatial scale.

The small (s) variant was selected over the nano (n) and medium (m) alternatives based on the A/B experimental results from Section 4.2. The nano model demonstrated a non-trivial false negative rate on partially occluded markers, which constitutes the critical failure mode for scale estimation, since a missed marker reduces the number of visible reference pairs. The medium model offered negligible improvement in recall over the small variant while approximately doubling inference time, making the small variant the optimal point on the accuracy-throughput trade-off curve for this application.

4.3.3 Refraction-Aware Scaling

As identified in preliminary experiments, frame-independent scaling is susceptible to refractive noise. To stabilize the scale, the system implements a Median Absolute Deviation (MAD) inlier mask [33]. This filter calculates the pixel-to-metre ratio (px/m) for all visible marker pairs and rejects outliers that deviate by more than k times the MAD from the median, where k was set to accommodate the expected variance in refraction-induced scale jitter while retaining the majority of valid observations. The MAD was preferred over the IQR for this inference-time filtering task because the number of visible marker pairs per frame can be very small, making the IQR either undefined or unreliable at such small sample sizes, whereas the MAD operates on the point-wise distribution of per-pair ratios and remains statistically valid even with a handful of observations. This ensures a stable calibration constant C for each frame.

When fewer than two markers are simultaneously visible, it is impossible to compute an inter-marker distance and therefore impossible to derive a fresh scale estimate. In this case, the most recent valid ratio ρ is carried forward to the current frame. This introduces a risk of temporal staleness: if the swimmer moves significantly in depth between calibration frames, the apparent size of the markers changes and the carried-forward ρ becomes inaccurate. This risk was accepted as a pragmatic trade-off given the typical geometry of poolside filming, where depth variation is small relative to the camera-to-lane distance and the resulting scale error is therefore bounded. The carry-forward mechanism is analogous to the zero-order hold in control systems and is a standard approach in tracking pipelines where sensor observations are intermittent.

4.4 Kinematic Metric Extraction

The final phase synthesizes the outputs of marker detection, ego-motion compensation, and pose estimation to produce absolute positional and kinematic metrics for the swimmer.

4.4.1 Ego-Motion Compensation

Because the camera is hand-held and moved manually alongside the swimmer, raw pixel coordinates are subject to both lateral drift and incidental shake between frames, making them incomparable across time without compensation. The choice of handheld video over a fixed camera installation was a deliberate architectural decision rather than a pragmatic compromise. A fixed camera would eliminate the need for ego-motion compensation entirely, but it would cover only a fraction of the lane from a single vantage point, requiring either multiple synchronized cameras or an acceptance that the majority of the swim would not be monitored. The handheld approach requires no permanent poolside infrastructure, making the system deployable at any competition or training venue without modification to the facility. The operator can pan or move the camera to keep the swimmer in frame throughout the full length of the pool, and it reflects the realistic data collection scenario facing a coaching practitioner without access to dedicated multi-camera installations.

To recover an absolute reference frame, the system exploits the lane markers as stationary anchors. For each consecutive frame pair, the horizontal pixel displacement of each detected marker is computed and matched to the nearest marker in the previous frame within a 50-pixel tolerance window. At the nominal pixel-to-metre ratio observed during trials, a 50-pixel window corresponds to a physical distance sufficient to accommodate the inter-frame camera displacement produced by typical handheld movement, while remaining small enough to prevent a marker in one position from being incorrectly matched to a different marker at the adjacent lane boundary. The median of all valid per-marker displacements gives a robust estimate of the camera's pixel shift Δx_{cam}, which is accumulated into a running global offset X_{cam}. The median is used in preference to the mean because the camera's motion is not uniform: brief sharp jolts from the operator's stride produce a non-Gaussian jitter distribution with heavy tails, for which the median is the appropriate central estimator:

$$X_{\text{cam}}^{(t)} = X_{\text{cam}}^{(t-1)} - \Delta x_{\text{cam}}^{(t)} \tag{4.2}$$

If no markers are detected in a given frame, the previous frame's displacement estimate is propagated forward (linear projection), preventing a discontinuity in the accumulated offset. The global pixel position of the swimmer is then:

$$p_{\text{global}} = p_{\text{pixel}} + X_{\text{cam}} \tag{4.3}$$

Where p_{pixel} is the swimmer's raw horizontal pixel coordinate in the current frame.

Feature-based optical flow, specifically the Lucas-Kanade tracker, was evaluated as a classical alternative for ego-motion estimation. However, the pool environment's repetitive textures generate numerous ambiguous feature matches that violate the spatial smoothness assumption underpinning optical flow methods. Electronic image stabilization, available in some consumer cameras, was considered as a hardware-level alternative but was unavailable on the recording hardware used in this project. The marker-based approach is semantically grounded in the geometry of the scene and is therefore more reliable in the repetitively textured aquatic environment than gradient-based feature matching.

4.4.2 Metric Scale Calibration

The pixel-to-metre conversion factor ρ (px/m) is estimated from the detected marker positions in each frame. Given N markers detected and sorted left-to-right, the outermost pair spans $N-1$ physical intervals of known separation d_{real} (metres). The scale is therefore:

$$\rho = \frac{\|\mathbf{p}_N - \mathbf{p}_1\|_2}{(N-1)\, d_{\text{real}}} \tag{4.4}$$

The estimate from the most recent frame in which two or more markers are visible is carried forward to frames where the marker count drops below two. The swimmer's absolute position in metres is then:

$$x_{\text{m}} = \frac{p_{\text{global}}}{\rho} \tag{4.5}$$

4.4.3 Swimmer Localization via ViTPose

The swimmer's position is localized using the ViTPose pose estimation model [8]. ViTPose was selected over two principal alternatives. The OpenPose bottom-up approach [17] was evaluated during preliminary testing but was found to fail frequently when a swimmer's limbs crossed the water-air boundary, as the part affinity field grouping mechanism relied on localized gradient features that are disrupted by refraction and surface glare at this interface. HRNet-based top-down models [34] were considered as a technically competitive alternative, however, ViTPose's plain Vision Transformer backbone [9] offers a specific advantage in the aquatic domain: its global self-attention mechanism aggregates contextual information from the full image at every layer, allowing the model to infer a joint's position from the broader kinematic context of the body even when that joint is locally indistinguishable from the water surface. This global receptive field is architecturally absent in convolutional HRNet variants, which rely on local convolutions and can be confused when localized keypoint evidence is obliterated by water.

Rather than using the bounding box centroid, the system extracts the left and right hip keypoints (COCO indices 11 and 12) and computes their midpoint as a proxy for the centre of mass (CoM):

$$\mathbf{c}_{\text{hip}} = \frac{\mathbf{k}_{11} + \mathbf{k}_{12}}{2} \tag{4.6}$$

The biomechanical rationale for this choice is that the centre of mass of a swimmer is located near the pelvis and hip region throughout all stroke phases. Using the bounding box centroid as a position proxy introduces a systematic error: during the arm entry phase of the stroke cycle, the extended arm shifts the bounding box centroid forward by a body-length fraction, introducing an artificial velocity oscillation at the stroke frequency that would contaminate the kinematic profile. The hip midpoint is anatomically stable relative to the true CoM across all stroke phases and is therefore a more appropriate proxy for a quantity intended to reflect whole-body translation. To account for any lateral offset between the swimmer and the marker baseline, the hip centroid is orthogonally projected onto the line segment connecting the outermost detected markers before the position is converted to metres.

Wrist keypoints (COCO indices 9 and 10) are additionally extracted with a confidence threshold of 0.3, providing left- and right-wrist positions in the global reference frame for downstream stroke-cycle analysis. The lower confidence threshold for wrists relative to hips reflects the deliberate asymmetry in their detectability: the hips are typically below the waterline and visible throughout the stroke in underwater video, while the wrists are frequently lifted out of the water, or occluded by splashes, causing ViTPose to assign lower confidence scores to anatomically valid detections. Retaining low-confidence wrist detections accepts a degree of additional noise in exchange for improved recall of true positive wrist positions, a trade-off that is appropriate because wrist data is used for qualitative stroke analysis rather than as the primary input to the position tracking pipeline.

4.4.4 Velocity and Kinematic Profile Extraction

The raw per-frame position records are first subjected to a Median Absolute Deviation (MAD) outlier filter and a temporal jump filter to remove implausible discontinuities. The MAD pre-filter is applied before the Kalman filter rather than relying on the filter alone because the Kalman filter's Gaussian noise model is not robust to large impulsive errors: a gross outlier, such as a pose estimation failure that places the hip keypoint at the edge of the frame, will corrupt the Kalman state estimate for several subsequent frames before the innovation sequence recovers. Removing such gross errors prior to ingestion into the filter preserves the integrity of the state estimate throughout the trajectory.

The cleaned trajectory is then processed by a discrete-time Kalman Filter [11], which models the swimmer as a constant-velocity system and simultaneously estimates position and instantaneous velocity from the noisy observations. The constant-velocity model is appropriate for the steady-state phase of a swim length, where accelerations between consecutive frames are small relative to the observation noise. It would be less appropriate at push-off or at the turning wall, where the swimmer undergoes rapid acceleration. A particle filter was considered as an alternative smoother for cases where the position distribution is multi-modal, but the swimmer's position distribution, conditioned on recent observations, is approximately Gaussian in the single-lane tracking scenario, making the Kalman filter optimal in the least-squares sense and computationally far less expensive. The process noise covariance Q encodes the expected acceleration variance between frames ($\sigma_a^2 = 3.0$ m^{2}s^{-4}), and the observation noise covariance R encodes the estimated metric-space position noise of the ViTPose hip detection after scale calibration ($\sigma_{\text{obs}}^2 = 0.015$ m^2).

Following Kalman filtering, a loop closure correction is applied to the estimated position track. Because scale calibration errors accumulate over the course of a length, the Kalman position estimate at the turning wall may not coincide precisely with the known pool dimension of 25 m. The correction rescales the outbound and return segments of the position track piecewise so that the peak of the trajectory aligns exactly with 25 m, enforcing physical consistency with the pool geometry. This step operates on the complete trajectory and is therefore a batch post-processing operation rather than a causal one as it cannot be performed until all frames have been processed.

Instantaneous velocity is then derived from the loop-closure-corrected position via numerical differentiation:

$$v(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{p_{\text{global}}(t)}{\rho}\right) \tag{4.7}$$

Numerical differentiation amplifies high-frequency noise in the position signal, necessitating a dedicated smoothing stage. A Savitzky-Golay filter is applied to the differentiated velocity signal, with a window length of 61 frames and a polynomial order of 3 for the hip centroid velocity. The Savitzky-Golay filter is well suited to this role precisely because, at this stage of the pipeline, all frames are available: unlike the Kalman filter, it is a batch operation and requires a symmetric window of neighbouring frames, which makes it inappropriate for causal position tracking but entirely appropriate for this final offline smoothing step. Its polynomial-fitting formulation preserves the shape of genuine kinematic events such as velocity peaks and stroke-cycle troughs more faithfully than a simple moving average, which would attenuate such features in proportion to their sharpness.

For wrist velocity signals, an extended cleaning pipeline is applied before the Savitzky-Golay smoothing stage: biological thresholding removes physically impossible velocity values, linear interpolation fills the resulting gaps produced by NaN-valued frames where the wrist keypoints were not detected, and a median filter suppresses residual impulse noise before the Savitzky-Golay filter is applied with a shorter window of 15 frames. The shorter window reflects the higher temporal frequency of the wrist motion relative to the whole-body translation of the hip centroid.

From the smoothed velocity estimates, summary kinematic metrics are derived: absolute speed profile, peak velocity, and mean velocity over the analysed swim segment.

4.5 Summary

This chapter presented a four-phase methodology for extracting swimmer kinematics from handheld poolside video, in which each phase was designed not in isolation but to address the specific error mode introduced by the preceding phase's output. The modular 3D-printed marker system was designed to remain visually identifiable under the refractive and turbulent conditions of the aquatic environment, with the alternating binary pattern and cylindrical geometry chosen to overcome the contrast and viewpoint-sensitivity failure modes that preclude the use of conventional planar calibration targets. Manual annotation was rejected as intractable at scale, synthetic generation as insufficiently realistic, and direct transfer learning as impractical in the absence of domain-matched public data. The Model-in-the-Loop pipeline, combining SAM 2 propagation with IQR-based outlier filtering and human verification, was therefore developed to construct a domain-specific training set at manageable cost. A refraction-aware MAD inlier filter was applied to stabilize the per-frame pixel-to-metre scale estimate derived from the known marker spacing, with the MAD selected over the IQR specifically because it remains valid when the number of observable reference pairs is small. Finally, ego-motion compensation using the markers as stationary anchors, ViTPose-based hip localization selected for its global attention mechanism and anatomical proximity to the centre of mass, and a Kalman filter smoothing stage were combined to produce absolute positional and velocity profiles from the raw tracking output. Feature-based optical flow was rejected for ego-motion estimation due to the repetitive textures of the pool environment, OpenPose was rejected due to its failure at the water-air boundary, and the particle filter was rejected on grounds of computational overhead given the approximately Gaussian position distribution. The Kalman filter handles causal position tracking, a loop closure step corrects accumulated scale error against the known 25 m pool geometry, and the Savitzky-Golay filter then smooths the velocity signal produced by numerical differentiation of the corrected trajectory, a role for which its non-causal polynomial formulation is well suited.

The system described in this chapter constitutes a complete pipeline from physical marker placement to calibrated kinematic output. Chapter 5 evaluates each component against quantitative accuracy benchmarks and characterizes the residual error sources that remain after the mitigation strategies described here have been applied.

5 Evaluation

5.1 Overview

This chapter evaluates the proof-of-concept system against quantitative ground truth data. Given the exploratory nature of the project, the evaluation focuses on a single representative swim sequence (participant P035, stroke breaststroke, pool length 25.0 m) in which the swimmer completes one full length and returns, yielding a 50 m trajectory. The primary metric is the temporal accuracy with which the system can recover the swimmer's position at known spatial checkpoints. Kinematic outputs, including velocity and wrist-relative motion profiles, are additionally presented and discussed.

5.2 Ground Truth Collection

Ground truth split times were collected retrospectively by reviewing the underwater video footage and manually recording the timestamp at which the swimmer's body crossed each 2.5 m marker boundary. This produced 10 outbound checkpoints (2.5 m to 25.0 m) and 10 return checkpoints (22.5 m back to 0.0 m), for a total of 20 reference observations. Times are expressed relative to the detected swim start, defined as the first frame in which sustained forward motion is observed. All splits were recorded by a single observer, thus no inter-rater reliability measure was applied, which represents an inherent limitation of the ground truth.

5.3 Marker Detection Accuracy

A foundational element of the positional tracking pipeline is the robust identification of the 2.5 m pool markers. To evaluate the generalizability of the trained object detection model, qualitative assessments were performed comparing its accuracy on the original training video's test split against an unseen video captured in the same pool but under varying environmental conditions.

Figure 5.1 presents this comparison. The left panels demonstrate the model's performance on test frames extracted from the same sequence used during training. Under these consistent conditions, the model exhibits high confidence and precise bounding box localization. The right panels depict the model's inference on the unseen sequence. While the physical pool geometry remains identical, differences in ambient lighting, water turbidity, surface glare, and camera positioning introduce a noticeable domain shift.

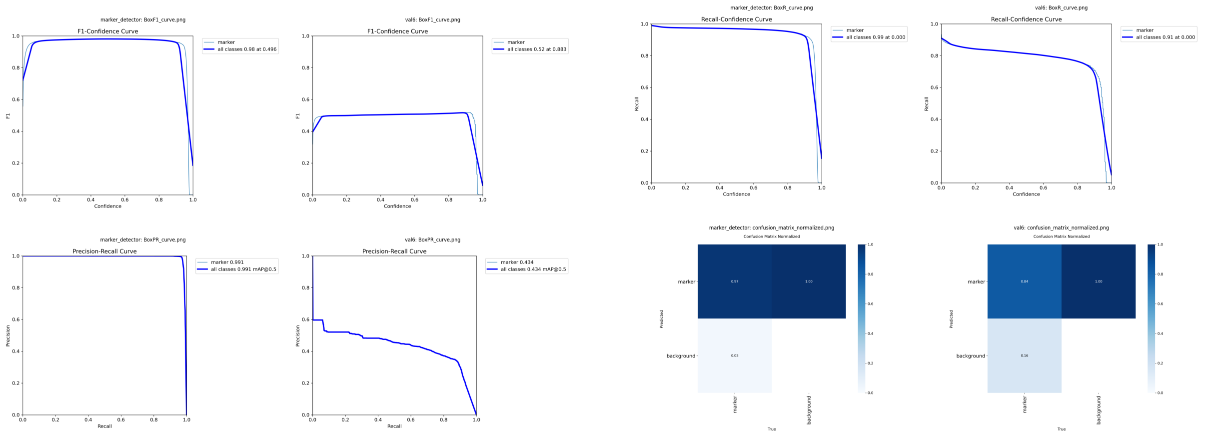

Figure 5.1 Comparison of marker detection accuracy. Left: Inference on test frames from the training video, showing high confidence and precision. Right: Inference on an unseen video from the same pool.

Consequently, the model experiences variations in confidence scores and occasional bounding box jitter in the unseen footage. This highlights a common limitation in underwater computer vision: high sensitivity to environmental fluctuations. These variations directly impact the stability of the frame-to-frame pixel-to-metre scale estimation. The subsequent application of the MAD-based inlier filter and Kalman smoothing (discussed in Section 5.4) is therefore critical to mitigate these detection inconsistencies and prevent them from propagating as severe positional drift.

5.4 Trajectory Optimization and Pre-Evaluation Processing

5.4.1 Outlier Removal

Prior to evaluation, the raw tracking records were subjected to the MAD-based inlier filter and temporal jump filter described in Chapter 4. Of the 4,052 raw records, 723 were flagged as significant deviations and removed, leaving 3,329 records for downstream processing. A secondary velocity ceiling of 5.0 m/s was subsequently applied, removing a further 32 records whose implied speed was physiologically implausible for competitive breaststroke swimming, justified by world-record 50 m breaststroke splits, which correspond to mean speeds of approximately 1.92 m/s [35]. Any inferred speed exceeding 5.0 m/s therefore constitutes an unambiguous tracking artefact.

5.4.2 Loop Closure Correction

The system detects the turn point as the frame index at which the Kalman-filtered position estimate reaches its maximum. For this sequence, the turn was detected at $t = 44.92$

s with a raw maximum position of 27.12 m, requiring an outbound scale correction factor of 0.9218 to anchor the peak exactly at 25.0 m. The return trajectory was subsequently re-mapped using a linear interpolation between the corrected peak and the expected return endpoint. Following loop closure, a Savitzky–Golay filter (window length 61, polynomial order 3) was applied to the centroid velocity signal to suppress residual high-frequency noise.

It should be noted that the two endpoint splits, 25.0 m on the outbound leg and 0.0 m on the return, could not be recovered and are reported as N/A in Table 5.1. The threshold-crossing detection searches for the first frame at which the trajectory strictly exceeds a target distance, since the trajectory is noisy near the turn and near the pool wall, it does not cleanly cross these boundary values, and no valid crossing index is found.

5.5 Positional Accuracy Evaluation

5.5.1 Split-Time Comparison

Table 5.1 presents the per-checkpoint comparison between the system's automatically derived split times and the manually recorded ground truth. The error is defined as:

$$e_i = t_{\text{auto},i} - t_{\text{true},i} \tag{5.1}$$

where a negative value indicates that the system reached checkpoint i earlier than the ground truth observation.

5.5.2 Analysis of Errors

The system achieves an overall MAE of 1.005 s and an RMSE of 1.210 s across 18 recoverable checkpoints. The error distribution exhibits distinct fluctuations rather than a uniform trend across phases. On the outbound leg, the system initially overestimates swimming speed in the mid-pool section, arriving at checkpoints early and reaching a peak error of -2.70 s at the 15.0 m mark. However, this trend reverses towards the end of the lap, where the system underestimates speed and arrives late, generating a $+2.02$ s error at 22.5 m. This suggests non-linear tracking distortions or fluctuating pixel-to-metre scale estimations rather than a simple accumulating drift. On the return leg, errors are slightly smaller in magnitude and more consistent, after initial late arrivals near the turn, the system reliably arrives slightly early, with errors clustering between -0.58 s and -1.48 s. The full trajectory comparison against ground truth checkpoints is shown in Figure 5.2, where oscillatory peaks are visible, reflecting the swimmer's imperfect stroke-cycle mechanics, exacerbated by the slight tracking errors.

Table 5.1 Updated per-checkpoint split time comparison: automated system vs. ground truth. Error $= t_{\text{auto}} - t_{\text{true}}$. (Start at index 385, Turn at index 2041)

Phase	Distance (m)	True Time (s)	Auto Time (s)	Error (s)
Outbound	2.5	2.00	2.12	$+0.12$
Outbound	5.0	4.00	3.88	-0.12
Outbound	7.5	7.00	5.37	-1.63
Outbound	10.0	9.00	8.75	-0.25
Outbound	12.5	12.00	10.15	-1.85
Outbound	15.0	14.00	11.30	-2.70
Outbound	17.5	17.00	16.03	-0.97
Outbound	20.0	20.00	20.50	$+0.50$
Outbound	22.5	22.00	24.02	$+2.02$
Outbound	25.0	26.00	N/A	N/A
Return	22.5	27.00	28.13	$+1.13$
Return	20.0	29.00	29.90	$+0.90$
Return	17.5	32.00	31.37	-0.63
Return	15.0	35.00	33.52	-1.48
Return	12.5	38.00	37.17	-0.83
Return	10.0	41.00	40.07	-0.93
Return	7.5	43.00	42.20	-0.80
Return	5.0	46.00	45.42	-0.58
Return	2.5	49.00	48.37	-0.63
Return	0.0	51.00	N/A	N/A
Overall MAE				**1.005 s**
Overall RMSE				**1.210 s**

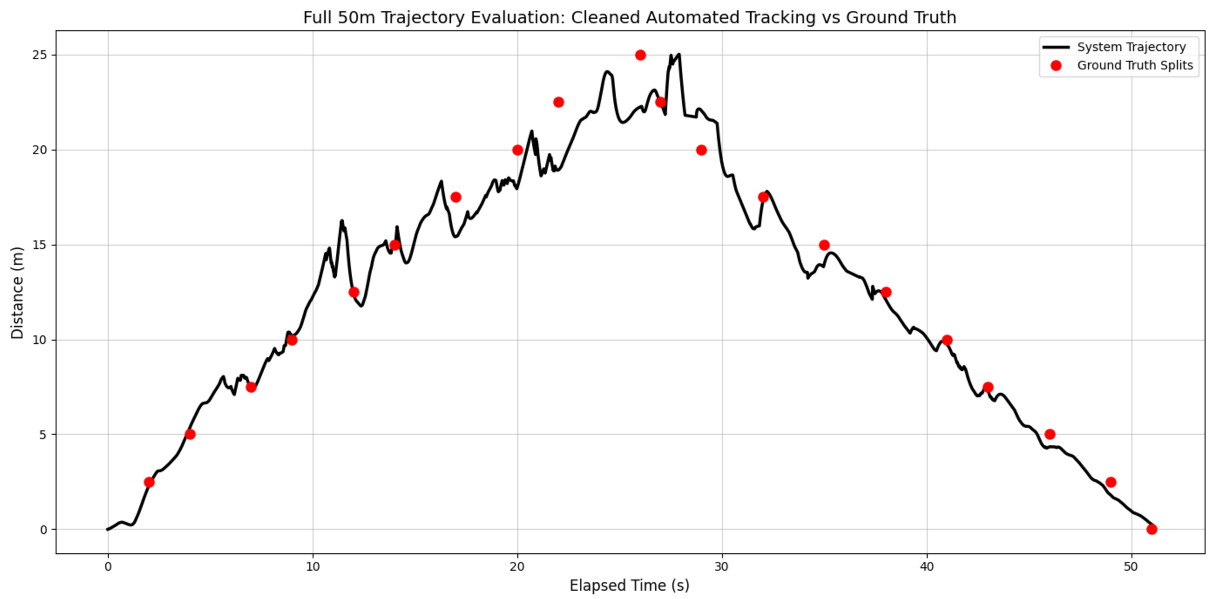

Figure 5.2 Full 50 m trajectory evaluation: system trajectory (black line) versus manually recorded ground truth split times (red points). The swimmer travels from 0 m to 25 m (outbound) and returns to 0 m.

5.6 Kalman Filter Noise Reduction

As an internal validation of the filtering pipeline, Table 5.2 reports the standard deviation of the raw and Kalman-smoothed estimates for both position and the pixel-to-metre scale factor. The Kalman filter achieves a modest reduction in position variance and a more substantial reduction in scale variance, reflecting its effectiveness at rejecting the frame-to-frame refractive noise that affects the marker-based scale estimate.

Table 5.2 Raw vs. Kalman-filtered standard deviation for position and scale estimates.

Signal	Raw σ	Kalman σ
Position (m)	7.4409	7.4430
Scale (px/m)	39.8377	37.6542

5.7 Kinematic Outputs

5.7.1 Centroid Velocity Profile

Following loop closure and smoothing, the swimmer's peak speed was estimated at 4.96 m/s (17.86 km/h) and the mean speed at 1.95 m/s (4.30 km/h). These are derived from the post velocity thresholding, as the pre velocity thresholding estimate yielded a higher peak of 6.97 m/s, which is physiologically implausible for competitive swimming and reflects residual noise prior to the final smoothing stage. The full velocity and speed profiles are shown in Figure 5.3.

5.7.2 Wrist Kinematics and Symmetry

To isolate the intra-stroke cycle mechanics from the swimmer's macroscopic forward translation, wrist kinematics were evaluated by subtracting the centroid position from the raw wrist coordinates. This centroid-anchored view is intended to reveal the cyclical pull and recovery phases relative to the swimmer's body. However, as visible in Figure 5.3, the resulting relative wrist signals exhibit substantial high-frequency noise and spike artefacts throughout the sequence, persisting despite biological thresholding, median filtering, and Savitzky–Golay smoothing. This instability highlights the limitations of the ViTPose keypoint predictions in challenging underwater environments, where partial occlusion, splash, and motion blur frequently cause distal keypoints like the wrists to be dropped or mislocated between frames.

Despite the inherent noise in individual trajectories, evaluating the coordination between limbs provides valuable insights. Because breaststroke dictates simultaneous

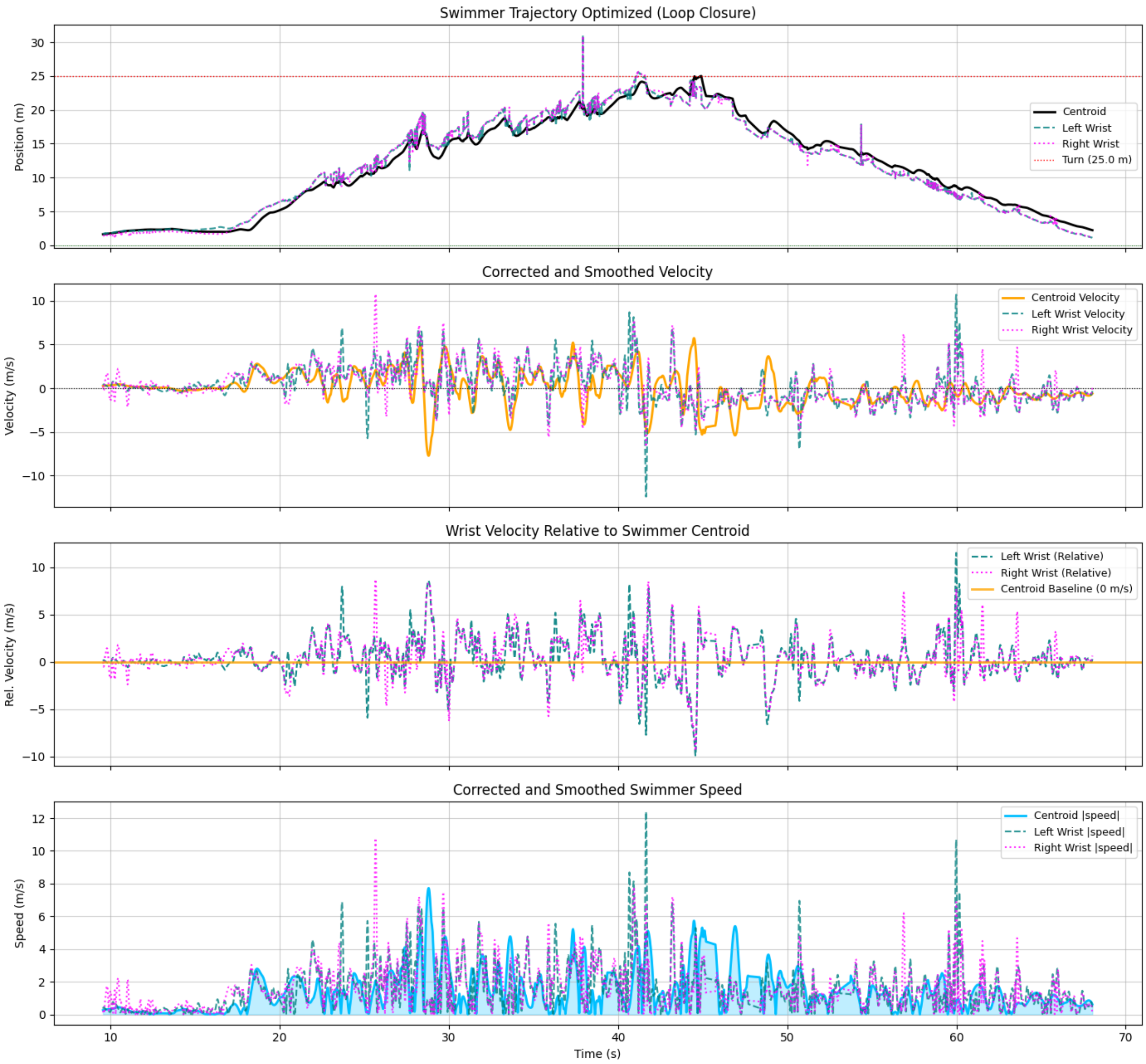

Figure 5.3 Kinematic profiles for the full 50 m sequence: (top) loop-closure-corrected trajectory for centroid and both wrists, (second) smoothed velocity, (third) wrist velocity relative to swimmer centroid, (bottom) absolute speed magnitude.

and symmetrical arm movements, the relationship between the left and right wrists was visualized using a heatmap (Figure 5.4), where both wrists should exhibit similar motion patterns. In an ideal stroke with perfect tracking, the distribution would manifest as a highly concentrated, overlapping cluster centred around the same relative position for both wrists. The observed heatmap, however, shows a more dispersed pattern with multiple density peaks, resulting from the noise from before the swim start, and from the shift in the swimmer's position for the turn, push-off and underwater glide phases.

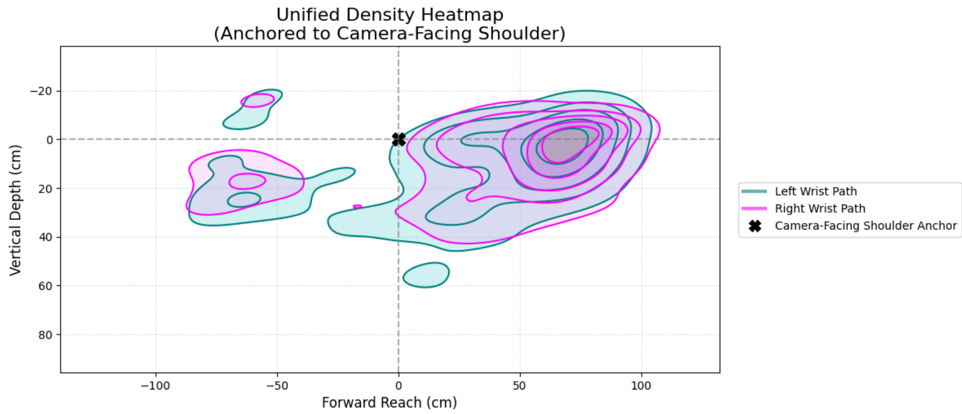

Figure 5.4 Symmetry heatmap illustrating the spatial correlation between the left and right wrist motions relative to the swimmer's centroid. High-density areas indicate the most common spatial relationship between the limbs during the stroke cycle.

The generated heatmap exhibits distinct central densities that indicate the underlying stroke rhythm and the overall synchronous nature of the breaststroke pull. However, the presence of asymmetrical dispersion and outliers in the heatmap further corroborates the tracking noise observed in the velocity profiles. While the macro-level phase correspondence between the arms is correctly captured by the model, remaining roughly in phase and appropriately positioned in front of the waist, the fine-grain kinematic noise limits the utility of the current system for precision technique analysis.

5.8 Summary

The system achieves a mean absolute split-time error of 1.005 s and an RMSE of 1.210 s across 18 checkpoints over a 50 m breaststroke sequence, demonstrating that marker-based ego-motion compensation and Kalman filtering can produce a usable positional trajectory from handheld, uncalibrated poolside footage. The primary positional limitation identified is the non-linear fluctuation in scale estimation on the outbound leg, resulting in alternating periods of overestimated and underestimated swimming speed rather than a strict, continuous monotonic drift. Distal keypoint tracking, particularly for the wrists, remains noisy in aquatic environments and represents the primary target for future refinement.

6 Conclusion

6.1 Summary of Contributions

This project demonstrates that end-to-end extraction of positional and kinematic metrics from handheld, uncalibrated poolside video is achievable using only open-source models and a low-cost 3D-printed physical marker, with no fixed poolside infrastructure required. Evaluated on a single 50 m breaststroke sequence (participant P035), the system recovered 18 of 20 spatial checkpoints with a mean absolute split-time error of 1.005 s and an RMSE of 1.210 s (Table 5.1). At the macro scale, the direction and phasing of the breaststroke arm cycle are recoverable from centroid-relative wrist signals; sub-stroke wrist kinematics, however, remain too noisy for reliable technique analysis under current conditions. The system is therefore best characterised as a proof-of-concept that validates the core approach and identifies the challenges that a production deployment would need to address, rather than a system ready for competitive coaching use.

6.2 Critical Reflection: Limitations and Trade-offs

The system's central value proposition is deployability: a coach can record a training session with a standard handheld camera and obtain split times and a velocity profile without installing any permanent poolside equipment. A fixed camera on a motorised rail, or a multi-camera rig with overlapping fields of view, would eliminate the need for ego-motion compensation entirely and would likely yield a substantially lower positional MAE. Such systems require venue-specific installation, a permanent power supply, and ongoing maintenance, placing them beyond the reach of most club-level coaching environments. The 1.005 s MAE achieved here is therefore best understood not as a deficiency relative to laboratory-grade systems, but as the cost of deployability. Whether this accuracy level is sufficient for practical coaching depends on the application: at a typical breaststroke speed of approximately 1.0 m/s, a 1 s timing error corresponds to roughly 1 m of positional uncertainty, which is adequate for lap timing and gross stroke-rate counting but insufficient for resolving inter-stroke speed modulations on the order of 0.1–0.2 m/s that are characteristic of detectable technique faults.

Beyond accuracy, the proof-of-concept scope introduces a significant generalisation risk. The fine-tuned YOLO26s detector was trained and evaluated on a single pool, swimmer, and lighting configuration; the MITL pseudo-labels encode the visual properties of that specific environment, and the detector inherits that specificity. Refraction introduces a systematic lateral bias in distal keypoint positions, particularly the wrists, that the current pipeline does not correct, as doing so would require knowledge of water depth and camera angle at every frame. ViTPose [8], pre-trained on the terrestrial COCO dataset [21], exhibits confidence collapse and occasional hallucination on submerged body parts where image statistics diverge sharply from its training distribution; this accounts for much of the high-frequency artefact observed in the wrist velocity signals. Transient occlusions from bubbles and surface glare can persist across multiple frames, exceeding the temporal scope of the MAD filter [33] and causing the scale estimate to carry forward stale values, while rapid camera panning during the swimmer's turn produces brief ego-motion estimation failures. Finally, the single-observer ground truth split times carry an unknown but non-zero perceptual timing uncertainty. Each of these limitations is individually manageable; together they imply that the reported MAE represents a lower bound on real-world deployment error rather than a performance ceiling.

6.3 Ethical and Practical Considerations

Video analysis of athletes raises immediate questions of privacy and consent. In this project, participation was governed by an institutional consent process; a deployed coaching tool would require explicit policies covering data retention periods, access controls, and athletes' rights to withdraw their data and request its deletion, none of which are determined by the technical architecture of the system.

Kinematic data derived from athlete footage constitutes performance intelligence with potential commercial value, relevant to talent identification, training optimisation, and competitive scouting. Whether such data belongs to the athlete, the coach, the club, or a technology provider is not resolved by how the pipeline is implemented; it requires explicit governance frameworks agreed upon before any deployment.

A further concern is the risk of over-reliance on numerical outputs. A system that produces split times and velocity profiles may lead coaches to treat those figures as ground truth rather than as noisy estimates subject to the failure modes described in Chapter 5. A deployable version of this system should surface uncertainty estimates or confidence intervals alongside its point predictions, making the limitations of the measurements legible to non-technical users and mitigating the risk of systematic over-interpretation.

6.4 Future Work

The most impactful near-term improvement to keypoint quality would be fine-tuning ViTPose [8] on an aquatic-domain dataset. The MITL pipeline developed in this project could be adapted to generate pseudo-labelled pose annotations at scale, creating a selfreinforcing data flywheel that progressively reduces the COCO distribution gap [21]. A domain-adapted model would directly address both the confidence collapse and the hallucination failures observed in wrist tracking, and would lay the groundwork for the sub-stroke technique analysis that the current system cannot reliably deliver.

Fusing the video-based trajectory with data from a waterproof inertial measurement unit worn by the swimmer offers a complementary sensing modality that is immune to optical refraction. IMU acceleration readings could serve as an additional observation in the Kalman filter [11], tightening the position estimate during marker-loss intervals and reducing dependence on the scale carry-forward heuristic that is a primary source of non-linear drift.

The current pipeline treats each frame largely independently, relying on the Kalman filter for inter-frame coherence. Replacing the YOLO-plus-Kalman tracking stack with a temporal transformer architecture, such as a video-based variant of SAM 2 [7] or a transformer-based multi-object tracker, would allow the model to reason over a window of learned temporal context. This could recover occluded marker positions and keypoints more robustly than the carry-forward heuristic and would reduce sensitivity to rapid camera motion during the turn.

Finally, the system currently operates as an offline post-processing pipeline. Adapting it for real-time inference would require profiling and optimising the YOLO26s [4] and ViTPose inference steps for GPU or NPU execution, and developing a coach-facing interface that presents split times, speed profiles, and stroke-symmetry indicators in a form accessible to non-technical users. Consistent with the considerations raised in Section 6.3, this interface should expose calibrated uncertainty estimates alongside point predictions rather than presenting a single authoritative number, ensuring that the system is used as a decision-support aid rather than as a source of unquestioned ground truth.

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Appendix A   Sample A

Appendix B   Sample B